do280-4.18: Days remaining **89** | Lab hours used **4/80**

DO280-4.22: Days remaining **89** | Lab hours used **0/80**

Red Hat OpenShift Administration II: Configuring a Production Cluster

Course | Lab Environment

Lab: Manage Application Security

Deploy an application that requires additional operating system privileges to run.

Deploy an application that requires access to the Kubernetes APIs to perform cluster maintenance tasks.

Outcomes

- Deploy a cluster maintenance application that must be executed regularly.
- Grant application access to Kubernetes APIs.
- Run an application with an SCC.

As the `student` user on the `workstation` machine, use the `lab` command to prepare your environment for this exercise, and to ensure that all required resources are available:

```
[student@workstation ~]$ lab start appsec-review
```

Instructions

In this exercise, you deploy two applications:

- A legacy payroll application, which must run as the fixed 0 UID to open the TCP 80 port.
- A project cleaner, which deletes projects with the `appsec-review-cleaner` label and that are longer than 10 seconds. This short expiration time is deliberate for the lab purposes.

You must deploy the project cleaner application to delete obsolete projects every minute.

The `lab start` command copies the required files for the exercise to the lab directory:

- A deployment manifest with the payroll application.
- A pod manifest that contains a project cleaner application. You can use this pod to test the project cleaner application and copy the pod specification into the cron job to complete the exercise.
- A manifest with the `project-cleaner` cluster role that grants the application access to find and delete namespaces.
- A cron job template file that you can edit to create cron jobs.
- A script that generates projects to verify that the project cleaner application works.

The API URL of your OpenShift cluster is `https://api.lab.example.com:6443`, and the `oc` command is already installed on your `workstation` machine.

Log in to the OpenShift cluster as the `developer` user with `developer` as the password. If you need cluster administration privileges, then log in as the `admin` user with `redhatocp` as the password.

1. Log in to your OpenShift cluster as the `developer` user and create the `appsec-review` project.

   1.1. Log in to the OpenShift cluster as the `developer` user with `developer` as the password.

   ```
   [student@workstation ~]$ oc login -u developer -p developer \
   https://api.lab.example.com:6443
   Login successful.

   ...output omitted...
   ```

   1.2. Create the `appsec-review` project.

   ```
   [student@workstation ~]$ oc new-project appsec-review
   Now using project "appsec-review" on server ...
   ...output omitted...
   ```

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2. Change to the ~/DO280/labs/appsec-review directory and deploy the payroll application in the payroll-app.yaml file. Verify that the application cannot run.

   2.1. Change to the ~/DO280/labs/appsec-review directory to access the lab files.

   ```
   [student@workstation ~]$ cd ~/DO280/labs/appsec-review
   ```

   2.2. Run the oc apply command to create the payroll deployment.

   ```
   [student@workstation appsec-review]$ oc apply -f payroll-app.yaml
   deployment.apps/payroll-api created
   ```

   2.3. Verify that the application fails to run by reading the deployment logs.

   ```
   [student@workstation appsec-review]$ oc logs deployment/payroll-api
   [2026-08-10 08:13:30 +0000] [1] [INFO] Starting gunicorn 20.1.0
   [2026-08-10 08:13:30 +0000] [1] [ERROR] Retrying in 1 second.
   [2026-08-10 08:13:31 +0000] [1] [ERROR] Retrying in 1 second.
   [2026-08-10 08:13:32 +0000] [1] [ERROR] Retrying in 1 second.
   [2026-08-10 08:13:33 +0000] [1] [ERROR] Retrying in 1 second.
   [2026-08-10 08:13:34 +0000] [1] [ERROR] Retrying in 1 second.
   [2026-08-10 08:13:35 +0000] [1] [ERROR] Can't connect to ('', 80)
   ```

   The container in the pod runs as root to listen on port 80.

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3. As the admin user, look for an SCC that allows the workload in the payroll-app.yaml deployment to run.

   3.1. Log in as the admin user with redhatocp as the password.

   ```
   [student@workstation appsec-review]$ oc login -u admin -p redhatocp \
   https://api.lab.example.com:6443
   Login successful.
   ...output omitted...
   ```

   3.2. Run the oc adm policy scc-subject-review command to get an SCC that allows the application to run.

   ```
   [student@workstation appsec-review]$ oc adm policy scc-subject-review
   \
   -f payroll-app.yaml
   RESOURCE                 ALLOWED BY
   Deployment/payroll-api   anyuid
   ```

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4. Create the `payroll-sa` service account and assign to it the SCC that the application requires. Then, assign the `payroll-sa` service account to the `payroll-api` deployment.

4.1. Run the `oc create` command to create the `payroll-sa` service account.

```
[student@workstation appsec-review]$ oc create sa payroll-sa
serviceaccount/payroll-sa created
```

4.2. Assign the `anyuid` SCC to the `payroll-sa` service account.

```
[student@workstation appsec-review]$ oc adm policy \
add-scc-to-user anyuid -z payroll-sa
clusterrole.rbac.authorization.k8s.io/system:openshift:scc:anyuid add
ed: "payroll-sa"
```

4.3. Use the `oc set serviceaccount` command to add the `payroll-sa` service account to the `payroll-api` deployment.

```
[student@workstation appsec-review]$ oc set serviceaccount deployment
\
payroll-api payroll-sa
deployment.apps/payroll-api serviceaccount updated
```

Hide Solution

5. Verify that the payroll API is accessible by running the `curl` command from the `payroll-api` deployment. Use the `http://localhost/payments/status` URL to verify that the API is working.

5.1. Use the `oc exec` command with the `payroll-api` deployment to run the `curl` command. Provide the `-sS` option to hide progress output and show errors.

```
[student@workstation appsec-review]$ oc exec deployment/payroll-api \
-- curl -sS http://localhost/payments/status
[{"id":240,"status":"Paid","userId":1003},{"id":241,"status":"Pendin
g","userId":1003}]
```

Hide Solution

6. Create the `project-cleaner-sa` service account and assign it to the `project-cleaner.yaml` pod manifest to configure the application permissions.

6.1. Create the `project-cleaner-sa` service account.

```
[student@workstation appsec-review]$ oc create sa project-cleaner-sa
serviceaccount/project-cleaner-sa created
```

6.2. Edit the `project-cleaner.yaml` pod manifest file and update the placeholder text to match the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: project-cleaner
  namespace: appsec-review
spec:
  restartPolicy: Never
  serviceAccountName: project-cleaner-sa
  containers:
    - name: project-cleaner
...output omitted...
```

Hide Solution

7. Create the `project-cleaner` role in the `cluster-role.yaml` file and assign it to the `project-cleaner-sa` service account.

7.1. Create the `project-cleaner` cluster role by applying the `cluster-role.yaml` manifest file.

```
[student@workstation appsec-review]$ oc apply -f cluster-role.yaml
clusterrole.rbac.authorization.k8s.io/project-cleaner created
```

7.2. Use the `oc adm policy add-cluster-role-to-user` command to add the `project-cleaner` role to the `project-cleaner-sa` service account.

```
[student@workstation appsec-review]$ oc adm policy add-cluster-role-to
-user \
project-cleaner -z project-cleaner-sa
clusterrole.rbac.authorization.k8s.io/project-cleaner added: "project
-cleaner-sa"
```

Hide Solution

8. Edit the `cron-job.yaml` file and update the placeholder text to create the `appsec-review-cleaner` cron job by using the `project-cleaner.yaml` pod manifest as the job template. Create the cron job and configure it to run every minute. You can use the solution file in the `~/DO280/solutions/appsec-review/cron-job.yaml` path.

   8.1. Edit the `cron-job.yaml` file to replace the `CHANGE_ME` string with the "`*/1 * * * *`" schedule to execute the job every minute.

   ```
   apiVersion: batch/v1
   kind: CronJob
   metadata:
     name: appsec-review-cleaner
     namespace: appsec-review
   spec:
     schedule: "*/1 * * * *"
     concurrencyPolicy: Forbid
     jobTemplate:
     ...output omitted...
   ```

   8.2. Replace the `CHANGE_ME` label in the `jobTemplate` definition with the `spec` definition from the `project-cleaner.yaml` pod manifest. Although the long image name might show across two lines, you must add it as one line.

   ```
   apiVersion: batch/v1
   kind: CronJob
   metadata:
     name: appsec-review-cleaner
     namespace: appsec-review
   spec:
     schedule: "*/1 * * * *"
     concurrencyPolicy: Forbid
     jobTemplate:
       spec:
         template:
           spec:
             restartPolicy: Never
             serviceAccountName: project-cleaner-sa
             containers:
               - name: project-cleaner
                 image: registry.lab.example.com:8443/redhattraining/do2
   80-project-cleaner:v1.1
                 imagePullPolicy: Always
                 env:
                 - name: PROJECT_TAG
                   value: "appsec-review-cleaner"
                 - name: EXPIRATION_SECONDS
                   value: "10"
   ```

   8.3. Create the cron job.

```
[student@workstation appsec-review]$ oc apply -f cron-job.yaml
cronjob.batch/appsec-review-cleaner created
```

Hide Solution

9. Optionally, verify that the project cleaner executed correctly. Use the `generate-projects.sh` script from the lab directory to generate projects for deletion. Wait for the next job execution and print the logs from that job's pod.

> **NOTE**
>
> The logs might not be in the last pod, but in the previous one.

9.1. Run the `generate-projects.sh` script to create test projects that the project cleaner will delete the next time that it runs.

```
[student@workstation appsec-review]$ ./generate-projects.sh
obsolete-appsec-review-1 created at 15:29:14
obsolete-appsec-review-2 created at 15:29:15
obsolete-appsec-review-3 created at 15:29:16
namespace/obsolete-appsec-review-1 labeled
namespace/obsolete-appsec-review-2 labeled
namespace/obsolete-appsec-review-3 labeled
Last appsec-review-cleaner label applied at 15:29:20
...output omitted...
```

9.2. List the pods in the `appsec-review` project until you see a pod with the `Completed` status that is later than the last label that the script applied.

```
[student@workstation appsec-review]$ oc get pods
NAME                                   READY   STATUS      RESTARTS
AGE
appsec-review-cleaner-27909204-g49gr   0/1     Completed   0
2m37s
appsec-review-cleaner-27909205-q2f2t   0/1     Completed   0
97s
appsec-review-cleaner-27909206-xcswb   0/1     Completed   0
37s
```

9.3. Print the logs from the last completed job, to verify that it deleted the obsolete projects.

```
[student@workstation appsec-review]$ oc logs pod/appsec-review-cleane
r-27909206-xcswb
...output omitted...
Namespace 'obsolete-appsec-review-1' deleted
Namespace 'obsolete-appsec-review-2' deleted
Namespace 'obsolete-appsec-review-3' deleted
```

9.4. Change to the home directory.

```
[student@workstation appsec-review]$ cd
```

Hide Solution

Evaluation

As the `student` user on the `workstation` machine, use the `lab` command to grade your work. Correct any reported failures and rerun the command until successful.

```
[student@workstation ~]$ lab grade appsec-review
```

Finish

As the `student` user on the `workstation` machine, use the `lab` command to complete this exercise. This step is important to ensure that resources from previous exercises do not impact upcoming exercises.

```
[student@workstation ~]$ lab finish appsec-review
```

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